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Engineering Notes

Nonslender Waverider Design Implementation into Simple Analytical Model for Waverider Design

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Nomenclature

A, B, C	=	power-law scaling parameters
D	=	drag
h	=	height
K_δ	=	hypersonic similarity parameter, $M\delta_c$
L	=	lift
l	=	length
M	=	Mach number
m, n	=	power-law exponents
t	=	thickness of base at symmetry plane
w	=	width
x, y, z	=	coordinates
α	=	angle of attack
β	=	shock inclination angle
γ	=	specific heat ratio
δ	=	leading-edge inclination angle
δ_c	=	cone angle
η	=	volumetric efficiency
Λ_b	=	waverider edge angle in base plane
θ	=	centerline wedge angle
τ	=	thickness ratio
ϕ_l	=	anhedral angle
Ω_∞	=	slope angle of freestream surface at edge in base plane

Subscripts

b	=	value for base surface
inv	=	inviscid
l	=	value of lower surfaces
u	=	value of upper surface
∞	=	freestream value

I. Introduction

THE simple analytical model for hypersonic waverider design of Starkey and Lewis [1] is a method that takes a power-law waverider and treats it as a series of wedges in which the shock or expansion waves are solved analytically. Some viscous effects are added as well. Several examples are solved in [1] and are compared with results based on more complex codes for waverider design; good agreement is found between the two. Further enhancements such as base pressure approximation and pitching moment calculation were added by Igra [2]. In [2], additional comparisons with both experimental and computational fluid dynamics (CFD) results of this extended method were presented, and good agreement was obtained with both.

One of the main difficulties with the simple analytical model for hypersonic waverider design is the design of variable wedge waveriders. This is unlike the constant wedge case in which the incident shock wave has a constant deflection angle from which all the design parameters can be derived. The variable wedge case, as its name suggests, is composed of several wedges that make up the waverider body. For example, if one knows the length, width, the center angle θ, and the leading-edge angle δ, either the power-law exponents n and m or the power-law scaling parameters (A, B, and C) are unknown. The current manuscript describes how to employ a nonslender waverider construction method as described in [3] and convert its design parameters into those of the simple analytical model for hypersonic waverider design. The nonslender waverider as described in [3] is based on a conical shock wave with various parameters affecting its geometry and angles of the leading edge and nose edge. By employing this waverider construction technique, it is possible to design a waverider using this method, to be converted into the simple analytical model for hypersonic waverider design of a variable wedge type. This conversion is not perfect, as some parameters of the simple analytical model for hypersonic waverider design should remain within certain bounds as discussed subsequently. However, its results are of similar quality with those shown in [1,2]. Next, a step-by-step description of the conversion process is presented, followed by comparison with CFD computations, in which good agreement was obtained.

II. Converting the Design

This section describes the process of taking the design parameters of the nonslender waverider of [3] and converting them to those of the simple analytical model for hypersonic waverider design of [1].

A. Centerline Angle θ

In [3], τ describes the relation between the waverider maximum center thickness and length

$$\tau = \frac{t}{\delta_c l} \tag{1}$$

where δ_c is the cone angle and measured in radians. Therefore, the centerline wedge angle is

$$\tan(\theta) = \frac{t}{l} = \tau\delta_c \tag{2}$$

In Fig. 1, a schematic description of the conical derived shock-wave flow and waverider location is presented. Based on the hypersonic small disturbance theory from [3], the shock deflection angle from the cone can be computed from the equation

$$\frac{\beta}{\delta_c} = \left(\frac{\gamma+1}{2} + \frac{1}{k_\delta^2}\right)^{1/2} \tag{3}$$

where $k_\delta = M_\infty\delta_c$.

B. Leading-Edge Inclination Angle δ

Turning our attention to Fig. 2, which describes the base of the waverider, it is possible to calculate h, which is the vertical distance of the tip of the waverider from its upper surface at the symmetry line.

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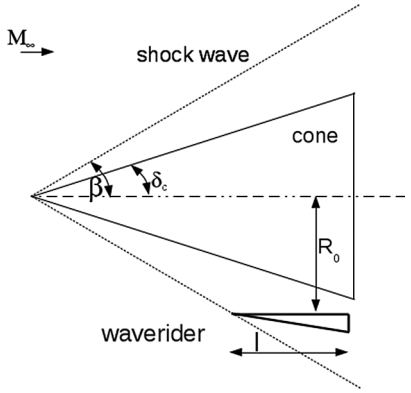


Fig. 1 A schematic description of the derived conical shock wave flow and the waverider.

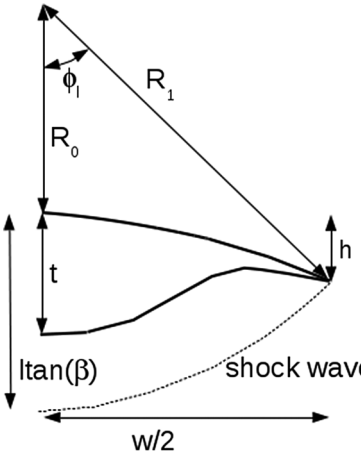


Fig. 2 Schematic description of the waverider base.

By using the following equations, the edge inclination angle δ is obtained:

$$R_1 = \frac{w}{2\sin(\phi_l)} \tag{4}$$

$$R_0 = R_1 - l\tan(\beta) \tag{5}$$

$$h = R_1\cos(\phi_l) - R_0 \tag{6}$$

This results in

$$h = R_1(\cos(\phi_l) - 1) + l\tan(\beta) \tag{7}$$

and finally, the leading-edge inclination angle δ is found:

$$\tan(\delta) = \frac{h}{l} \tag{8}$$

C. Computing Power-Law Exponents *m* and *n*

The angles shown in Fig. 3 are the base angles of the nonslender waverider of [3]. These will be converted so as to be suitable for use with the equations of (y_l and y_u) for the simple analytical model for hypersonic waverider design shown in Fig. 4. The locations of the centerline wedge angle and the anhedral angle are shown in this figure as well.

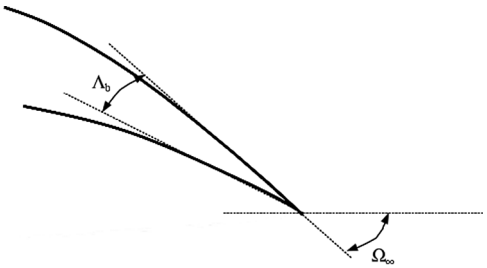


Fig. 3 Schematic description of the waverider base angles.

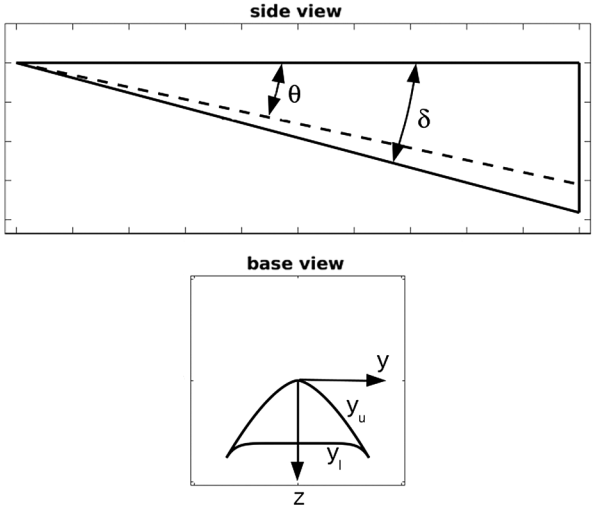


Fig. 4 Schematic description of the waverider base's coordinate system.

First, we turn our attention to the upper line marked y_u in Fig. 4 and defined by

$$y_u = Bz^n \tag{9}$$

the derivative of which is

$$\frac{dy_u}{dz} = Bnz^{n-1} \tag{10}$$

which at the tip equals $\tan(\Omega_\infty)$; therefore at the tip, where $y = w/2$ and $z = h$,

$$n = \frac{2h}{w}\tan(\Omega_\infty) \tag{11}$$

Next, we examine the lower curve y_l, which is defined by

$$y_l = C(l\tan(\theta) - z)^m \tag{12}$$

the derivative of which is

$$\frac{dy_l}{dz} = -Cm(l\tan(\theta) - z)^{m-1} \tag{13}$$

At the tip, this derivative is equal to $\tan(\Omega_\infty - \Lambda_b)$, thus obtaining m:

$$m = \frac{-2(l\tan(\theta) - h)\tan(\Omega_\infty - \Lambda_b)}{w} \tag{14}$$

However, in some cases, the results for m of Eq. (14) might be too small; in such a case, the lower surface of the waverider intersects the upper surface creating an unphysical design. Equation (50) in [1] defines the minimum value of m so that this does not occur; this minimum value is defined in the following equation:

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$$m > \frac{n(\tan(\delta) - \tan(\theta))}{\tan(\delta)} \tag{15}$$

Therefore, the value of m computed by Eq. (14) must be larger than that of m computed by Eq. (15); if not, the result of Eq. (15) is implemented in the waverider design. This is the only limitation of this conversion.

With this, the conversion from the nonslender waverider parameters into the simple analytical model for the hypersonic waverider design method is completed. Once this stage is reached, the waverider can be computed analytically by the method described in [1] with the extensions of [2], and its aerodynamic coefficients are obtained.

III. Results

Now that the method for converting the nonslender waverider design has been completed, an example of this conversion will be presented. For this, case 1 from [3] was chosen. This waverider has a length of 20 m flying at Mach number 8 at an altitude of 24 km, exactly as presented in [3]. At these conditions, the Reynolds number based on the length of the waverider is 1.55×10^8. In [3], this waverider was computed with the hypersonic small disturbance theory with a laminar boundary layer approximation; see [3] for further details. The design parameters taken from [3] are as follows: $M_\infty = 8, \delta_c = 10$ deg, $\Omega_\infty = 30$ deg, $\Lambda_b = 24.3$ deg, $\tau = 0.888$, and $\phi_l = 27.2$ deg. Once the conversion is complete, the new parameters compatible with the simple analytical model for hypersonic waverider design are as follows: $\theta = 8.81$ deg, $\delta = 10.8$ deg, $n = 0.641$, $m = 0.12$, and $w = 6.88$ m. In this case, m was computed with the minimum value given by Eq. (15). This waverider is shown in Fig. 5. It has a volumetric efficiency of 0.236 compared with 0.239 for the geometry of [3]. The overall shape of this waverider appears to be similar to that shown in Fig. 5 of [3].

Table 1 Summary of aerodynamic coefficients

Results	L/D	L/D_{inv}	C_L	C_D	$C_{D\text{inv}}$
From [3]	7	7.26	0.0668	0.00955	0.00921
Current $p_b = p_\infty$	5.82	6.99	0.0649	0.0111	0.0093
Current $p_b = f(M_\infty)$	4.74	5.48	0.0649	0.0137	0.0117

A summary of the aerodynamic coefficients is presented in Table 1 for the current results computed by simple analytical model for hypersonic waverider design and as compared with those of [3]. In both cases, the reference area is the planform area, which is 83.87 m^2.

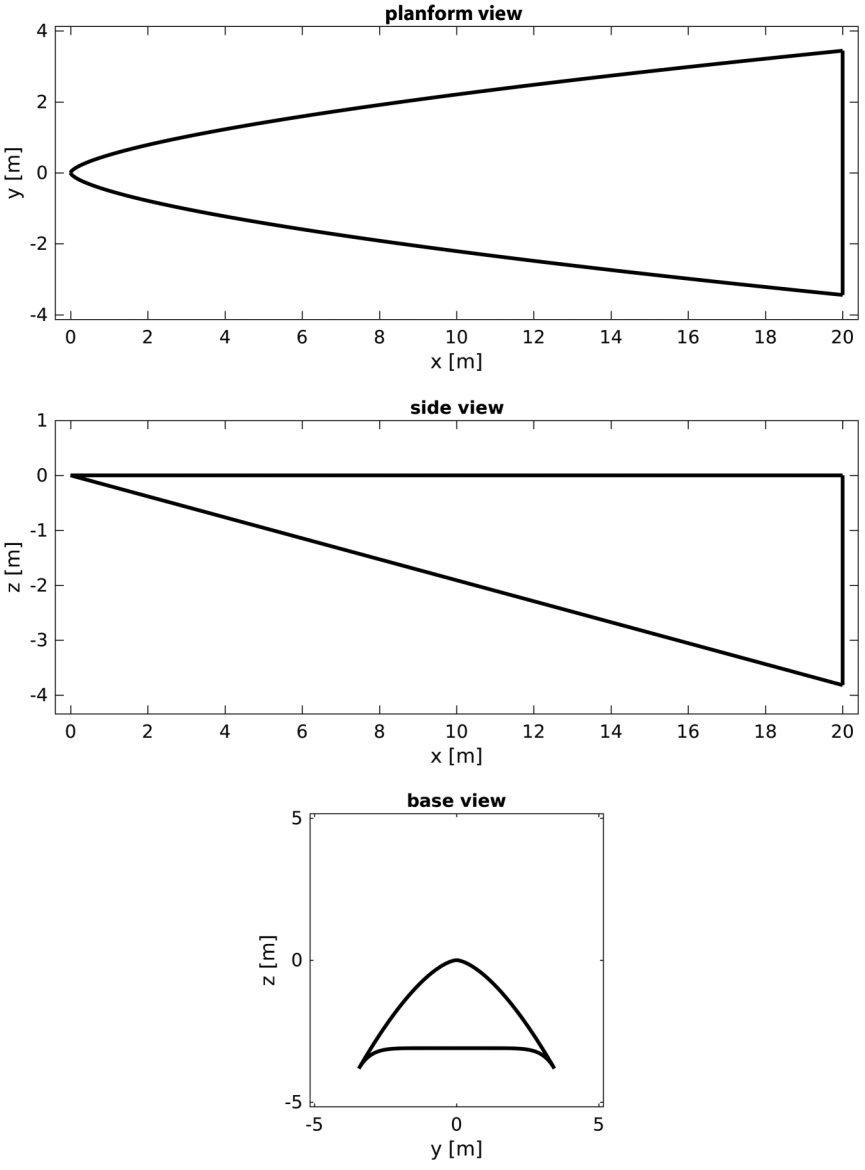


Fig. 5 Description of the waverider after the conversion.

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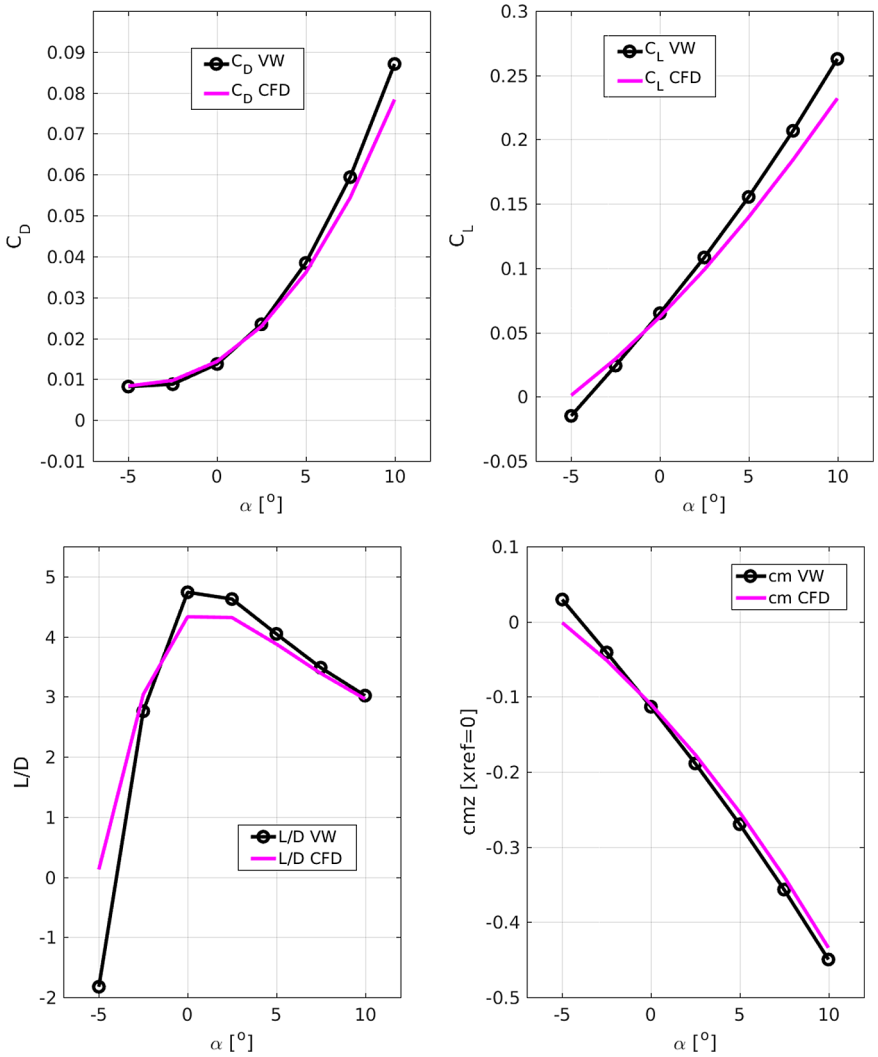


Fig. 6 Comparison between the aerodynamic coefficients obtained by the analytical method and CFD (VW, variable wedge).

The current results were computed for two cases, once with $p_b = p_\infty$ as done in [3] and once with $p_b = f(M_\infty)$ as described in [2]. Comparing the inviscid results of the two methods when $p_b = p_\infty$, we note that they are similar. However, the results of [3] are overly optimistic regarding the drag and L/D values for real applications. This stems from two reasons. The first is that in [3] the viscous effects were considered laminar despite the large Reynolds number that suggests a turbulent flow regime. The second is the approximation of the base pressure of which the effects on the drag and L/D were shown in [2]. Based on these, the L/D drops from 7 (taken from [3]) to 4.74 in the current computation, a drop of over 30%. To verify the current results, a CFD computation was conducted for the configuration shown in Fig. 5 with the addition of very small rounded leading edges. It was solved by a second-order compressible flow solver with a Spalart–Allmaras turbulence model. The mesh for half a geometry (based on left–right symmetry) contained 4×10^6 cells. The results of this comparison are shown in Fig. 6 for angle of attack from -5 to 10 deg. Overall, there appears to be good agreement between the numerical and the analytical results. This is quite surprising, considering that the latter is based on a simple analytic code with a slightly modified geometry. The value of L/D at zero angle of attack in the CFD is 4.33, which is quite close (within 10%) to the analytic result of 4.74 especially when compared with the results from [3].

IV. Conclusions

A method for designing a nonslender waverider was added to the simple analytical model for hypersonic waverider design. An example of a nonslender waverider conversion was presented. This waverider was solved with the simple analytical model for hypersonic waverider design and computational fluid dynamics (CFD). Good agreement with CFD results was obtained especially when considering the results from [3], which did not take into account the correct viscosity and base pressure.

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